

an implicit solution of the first equation: $x = -y^2 - 2y - 2 + ce^y$.

(ii) Because mathematicians thought they were appropriately descriptive, certain words were “adopted” from engineering and made their own. The word *transient*, used earlier, is one of these terms. In future discussions the words *input* and *output* will occasionally pop up. The function f in (2) is called the **input** or **driving function**; a solution of the differential equation for a given input is called the **output** or **response**.

(iii) The term *special functions* mentioned in conjunction with the error function on page 58 also applies to the sine integral function and the Fresnel sine integral function introduced in Problems 47 and 48 in Exercises 2.3. “Special Functions” is actually a well-defined field of study in mathematics. The Fresnel sine integral function is one of two such special integral functions that the French engineer/physicist **Augustin-Jean Fresnel** (1788–1827) encountered in his study of optics. More special functions are studied in Section 5.3.

2.3 Exercises

Answers to selected odd-numbered problems begin on page ANS-2.

In Problems 1–24, find the general solution of the given differential equation. Give the largest interval over which the general solution is defined. Determine whether there are any transient terms in the general solution.

1. $\frac{dy}{dx} = 5y$
2. $\frac{dy}{dx} + 2y = 0$
3. $\frac{dy}{dx} + y = e^{3x}$
4. $3\frac{dy}{dx} + 12y = 4$
5. $y' + 3x^2y = x^2$
6. $y' + 2xy = x^3$
7. $x^2y' + xy = 1$
8. $y' = 2y + x^2 + 5$

9. $x \frac{dy}{dx} - y = x^2 \sin x$
10. $x \frac{dy}{dx} + 2y = 3$
11. $x \frac{dy}{dx} + 4y = x^3 - x$
12. $(1 + x) \frac{dy}{dx} - xy = x + x^2$
13. $x^2 y' + x(x + 2)y = e^x$
14. $xy' + (1 + x)y = e^{-x} \sin 2x$
15. $ydx - 4(x + y^6)dy = 0$
16. $ydx = (ye^y - 2x)dy$
17. $\cos x \frac{dy}{dx} + (\sin x)y = 1$
18. $\cos^2 x \sin x \frac{dy}{dx} + (\cos^3 x)y = 1$
19. $(x + 1) \frac{dy}{dx} + (x + 2)y = 2xe^{-x}$
20. $(x + 2)^2 \frac{dy}{dx} = 5 - 8y - 4xy$
21. $\frac{dr}{d\theta} + r \sec \theta = \cos \theta$
22. $\frac{dP}{dt} + 2tP = P + 4t - 2$
23. $x \frac{dy}{dx} + (3x + 1)y = e^{-3x}$
24. $(x^2 - 1) \frac{dy}{dx} + 2y = (x + 1)^2$

In Problems 25–32, solve the given initial-value problem. Give the largest interval I over which the solution is defined.

25. $xy' + y = e^x, y(1) = 2$
26. $y \frac{dx}{dy} - x = 2y^2, y(1) = 5$
27. $L \frac{di}{dt} + Ri = E; i(0) = i_0, L, R, E, \text{ and } i_0 \text{ constants}$

28. $\frac{dT}{dt} = k(T - T_m); T(0) = T_0, K, T_m, \text{ and } T_0 \text{ constants}$

29. $(x + 1) \frac{dy}{dx} + y = \ln x, y(1) = 10$

30. $y' + (\tan x)y = \cos^2 x, y(0) = -1$

31. $\left(\frac{e^{-2\sqrt{x}} - y}{\sqrt{x}} \right) \frac{dx}{dy} = 1, y(1) = 1$

32. $(1 + t^2) \frac{dx}{dt} + x = \tan^{-1} t, x(0) = 4$

[Hint: In your solution let $u = \tan^{-1} t$.]

In Problems 33–36, proceed as in Example 5 to solve the given initial-value problem. Use a graphing utility to graph the continuous function $y(x)$.

33. $\frac{dy}{dx} + 2y = f(x), y(0) = 0, \text{ where}$

$$f(x) = \begin{cases} 1, & 0 \leq x \leq 3 \\ 0, & x > 3 \end{cases}$$

34. $\frac{dy}{dx} + y = f(x), y(0) = 1, \text{ where}$

$$f(x) = \begin{cases} 1, & 0 \leq x \leq 1 \\ -1, & x > 1 \end{cases}$$

35. $\frac{dy}{dx} + 2xy = f(x), y(0) = 2, \text{ where}$

$$f(x) = \begin{cases} x, & 0 \leq x < 1 \\ 0, & x \geq 1 \end{cases}$$

36. $(1 + x^2) \frac{dy}{dx} + 2xy = f(x), y(0) = 0, \text{ where}$

$$f(x) = \begin{cases} x, & 0 \leq x < 1 \\ -x, & x \geq 1 \end{cases}$$

In Problems 37 and 38, proceed as in Example 5 to solve the given initial-value problem. Use a graphing utility to graph the continuous function $y(x)$.

37. $\frac{dy}{dx} + P(x)y = 4x, \quad y(0) = 3, \text{ where}$

$$P(x) = \begin{cases} 2, & 0 \leq x \leq 1 \\ -\frac{2}{x}, & x > 1 \end{cases}$$

38. $\frac{dy}{dx} + P(x)y = 0, \quad y(0) = 4, \text{ where}$

$$P(x) = \begin{cases} 1, & 0 \leq x \leq 2 \\ 5, & x > 2 \end{cases}$$

In Problems 39 and 40, the given differential equation is not linear in y . Nevertheless, find a general solution of the equation.

39. $dx = (x + y^2) dy$

40. $ydx + (2x + xy - 3) dy = 0$

In Problems 41 and 42, proceed as in Example 6 and express the solution of the given initial-value problem in terms of $\text{erf}(x)$ (Problem 41) and $\text{erfc}(x)$ (Problem 42).

41. $\frac{dy}{dx} - 2xy = 1, \quad y(1) = 1$

42. $\frac{dy}{dx} - 2xy = 1, \quad y(0) = \sqrt{\pi}/2$

In Problems 43–46, proceed as in Example 6 and express the solution of the given initial-value problem in terms of an integral-defined function.

43. $\frac{dy}{dx} + e^x y = 1, \quad y(0) = 1$

44. $x^2 \frac{dy}{dx} - y = x^3, \quad y(1) = 0$

45. $\frac{dy}{dx} + xy = \cos x, \quad y(0) = 2$
46. $\frac{dy}{dx} + (\sin x)y = 10x, \quad y(0) = 0$
47. The **sine integral function** is defined as

$$\text{Si}(x) = \int_0^x \frac{\sin t}{t} dt,$$

where the integrand is defined to be 1 at $x = 0$. Express the solution of the initial-value problem

$$x^3 \frac{dy}{dx} + 2x^2 y = 10 \sin x, \quad y(1) = 0$$

in terms of $\text{Si}(x)$.

48. The **Fresnel sine integral function** is defined as

$$S(x) = \int_0^x \sin\left(\frac{\pi}{2} t^2\right) dt.$$

Express the solution of the initial-value problem

$$\frac{dy}{dx} - (\sin x^2)y = 0, \quad y(0) = 5$$

in terms of $S(x)$.

Discussion Problems

49. Reread the discussion following Example 1. Construct a linear first-order differential equation for which all nonconstant solutions approach the horizontal asymptote $y = 4$ as $x \rightarrow \infty$.
50. Reread Example 2 and then discuss, with reference to Theorem 1.2.1, the existence and uniqueness of a solution of the initial-value problem consisting of $xy' - 4y = x^6 e^x$ and the given initial condition.

(a) $y(0) = 0$

(b) $y(0) = y_0, \quad y_0 > 0$

(c) $y(x_0) = y_0, \quad x_0 > 0, \quad y_0 > 0$

51. Reread Example 3 and then find the general solution of the differential equation on the interval $(-3, 3)$.
52. Reread the discussion following Example 4. Construct a linear first-order differential equation for which all solutions are asymptotic to the line $y = 3x - 5$ as $x \rightarrow \infty$.
53. Reread Example 5 and then discuss why it is technically incorrect to say that the function in (13) is a *solution* of the IVP on the interval $[0, \infty)$.
54. (a) Construct a linear first-order differential equation of the form $xy' + a_0(x)y = g(x)$ for which $y_c = c/x^3$ and $y_p = x^3$. Give an interval on which $y = x^3 + c/x^3$ is the general solution of the DE.
- (b) Give an initial condition $y(x_0) = y_0$ for the DE found in part (a) so that the solution of the IVP is $y = x^3 - 1/x^3$. Repeat if the solution is $y = x^3 + 2/x^3$. Give an interval I of definition of each of these solutions. Graph the solution curves. Is there an initial-value problem whose solution is defined on the interval $(-\infty, \infty)$?
- (c) Is each IVP found in part (b) unique? That is, can there be more than one IVP for which, say, $y = x^3 - 1/x^3$, x in some interval I is the solution?
55. In determining the integrating factor (5), there is no need to use a constant of integration in the evaluation of $\int P(x) dx$. Explain why using $\int P(x) dx + c$ has no effect on the solution of (2).

Mathematical Models

56. **Radioactive Decay Series** The following system of

differential equations is encountered in the study of the decay of a special type of radioactive series of elements:

$$\begin{aligned}\frac{dx}{dt} &= -\lambda_1 x, \\ \frac{dy}{dt} &= \lambda_1 x - \lambda_2 y,\end{aligned}$$

where λ_1 and λ_2 are constants. Discuss how to solve this system subject to $x(0) = x_0$, $y(0) = y_0$. Carry out your ideas.

- 57. Heart Pacemaker** A heart pacemaker consists of a switch, a battery of constant voltage E_0 , a capacitor with constant capacitance C , and the heart as a resistor with constant resistance R . When the switch is closed, the capacitor charges; when the switch is open, the capacitor discharges, sending an electrical stimulus to the heart. During the time the heart is being stimulated, the voltage E across the heart satisfies the linear differential equation

$$\frac{dE}{dt} = -\frac{1}{RC}E.$$

Solve the DE subject to $E(4) = E_0$.

Computer Lab Assignments

- 58. (a)** Use a CAS to graph the solution curve of the initial-value problem in Problem 42 on the interval $(-\infty, \infty)$.
(b) Use tables or a CAS to find the value $y(2)$.
- 59. (a)** Use a CAS to graph the solution curve of the initial-value problem in Problem 47 on the interval $[0, \infty)$.
(b) Use a CAS to find the value of the absolute maximum of the solution $y(x)$ on the interval.
- 60. (a)** Use a CAS to graph the solution curve of the initial-value problem in Problem 48 on the interval $(-\infty, \infty)$.

(b) It is known that Fresnel sine integral $S(x) \rightarrow \frac{1}{2}$ as $x \rightarrow \infty$ and $S(x) \rightarrow -\frac{1}{2}$ as $x \rightarrow -\infty$. What does the solution $y(x)$ approach as $x \rightarrow \infty$? As $x \rightarrow -\infty$?

(c) Use a CAS to find the values of the absolute maximum and the absolute minimum of the solution $y(x)$ on the interval.